

Wind Farm Project Finance - Methods Note

How the model connects operating assumptions to credit-analysis outputs

Workflow

- Start with operating assumptions: MWh sold, price per MWh, staffing, operating costs, depreciation, and tax assumptions.
- Build operating performance: revenue, operating costs, EBITDA, depreciation, EBIT, taxes, and cash flow before financials.
- Layer in capital structure: senior debt, mezzanine capital, sponsor equity, additional sponsor support, interest rates, and principal repayment schedule.
- Construct the cash-flow waterfall: debt service, repayment reserve, liquidity reserve, dividends, and terminal reserve release.
- Evaluate covenants: ADSCR, LLCR, ICR, and equity/mezzanine capital requirements.
- Run stress cases: lower MWh production and lower later-period power prices.

Coverage ratios

Ratio	Portfolio interpretation
ADSCR	Annual cash flow capacity relative to annual debt service. The project needed to remain at or above 1.60.
LLCR	Discounted remaining cash-flow support relative to remaining debt. The project needed to remain at or above 1.90.
ICR	Ability to cover interest expense from project cash flow. The project needed to remain at or above 1.70.

Modeling note

The public workbook is a cleaned demonstration artifact. It preserves the core project-finance logic and scenario outputs while avoiding the complexity of the full class template. It is intended for portfolio review by recruiters, graduate admissions readers, and finance-oriented reviewers.